

Targeted Lobbying on Council Networks

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1. Motivation and Key Questions

The classical model of political corruption (Shleifer and Vishny, 1993) treats public officials as isolated agents whose decisions to accept bribes are independent. Real legislative bodies are densely interconnected through parties, committees, and ideology. A lobbyist with a limited budget must therefore decide not only *how hard* to push but *whom* to push, knowing that influencing one well-connected member can cascade into broader support. We ask: (i) how does the network structure of the council determine the lobbyist’s optimal targeting strategy; (ii) how does the voting rule (simple majority, supermajority, count-based) interact with that structure to shape lobbying intensity and concentration; and (iii) can the choice of voting rule reduce the welfare cost of lobbying without restricting communication among legislators?

2. Model (Baseline)

We study a static, one-shot game in the linear–quadratic network-game tradition of Galeotti, Golub, and Goyal (2020). A lobbyist faces a council of n members $i \in \{1, \dots, n\}$ connected by an undirected weighted network $G = (g_{ij})$ with $g_{ii} = 0$. Each member chooses a continuous level of support $a_i \in \mathbb{R}$ —interpreted as effort toward passage (floor speeches, whipping colleagues, committee concessions)—to maximize

$$U_i = a_i \left(b_i + t_i + \beta \sum_j g_{ij} a_j \right) - \frac{1}{2} c a_i^2,$$

where $b_i \in \mathbb{R}$ is i ’s baseline inclination, $t_i \geq 0$ is the lobbyist’s targeted effort on i , $\beta > 0$ governs peer-effect strength, and $c > 0$ captures the convex cost of visible support. Following GGG, we leave a_i unconstrained in $\mathbb{R}$; concavity in own action plus the spectral condition below yield a finite equilibrium automatically. The lobbyist commits to $t = (t_1, \dots, t_n) \geq 0$, pays total cost $\sum_i t_i$, and the bill passes iff $\sum_i a_i \geq \bar{T}$. As an alternative specification, we are considering replacing the quadratic cost with an entropy term $\kappa[a_i \log a_i + (1 - a_i) \log(1 - a_i)]$, whose FOC gives a sigmoid best response $a_i^* = \sigma((b_i + t_i + \beta \sum_j g_{ij} a_j)/\kappa)$, mapping actions naturally into $(0, 1)$ at the cost of a closed-form equilibrium.

Equilibrium. Stacking the first-order conditions yields a linear best-response system. Under the spectral condition $c > \beta \lambda_{\max}(G)$, the council subgame has a unique Nash equilibrium $a^*(t) = (cI - \beta G)^{-1}(b + t)$. Let $M := (cI - \beta G)^{-1}$ and define the *Katz–Bonacich centrality vector with decay* β/c as $v := M\mathbf{1}$. Total equilibrium support is then $\mathbf{1}^\top a^*(t) = v^\top b + v^\top t$, and the lobbyist’s cost-minimization problem reduces to the linear program $\min_{t \geq 0} \mathbf{1}^\top t$ subject to $v^\top t \geq \bar{T} - v^\top b$.

3. Conjectures, Implications, and Plan

Our central conjecture is that under simple majority, the lobbyist optimally concentrates the entire targeting budget on a single member with the highest Katz–Bonacich centrality, paying minimum cost $(\bar{T} - v^\top b)/\max_i v_i$. This refines the eigenvector-centrality result of GGG, which emerges only in the near-critical limit $\beta \lambda_{\max}(G) \rightarrow c$. The main contribution we aim for concerns how this rule responds to the voting rule. As $\bar{T}$ rises from simple majority to supermajority and unanimity, we expect the optimal strategy to shift from *concentration* on the most central member to *spreading* effort across many, with count-based rules breaking linearity and forcing multi-member targeting. The cost saving from correct targeting should be increasing in β/c and in the heterogeneity of centralities, so hierarchical networks are cheapest to lobby and flat regular networks most resistant—paralleling Morris (2000)’s contagion analysis.

These results imply that network position should predict who is targeted by organized interests, and that voting rules and network structure act as *substitutes* in resisting capture: a supermajority rule in a flat network offers protection comparable to simple majority in a hierarchical one. Reforms strengthening peer ties among legislators may therefore unintentionally raise the chamber’s vulnerability to concentrated lobbying. We plan to prove the baseline result, extend the LP to supermajority and count-based rules, compute the centrality premium on canonical network families, and develop one extension—directed networks, competing lobbyists, or a noisy electoral signal linking back to Shleifer–Vishny—as our contribution beyond GGG.